SECTION 15060

PIPE AND PIPE FITTINGS

## PART 1 GENERAL

### 1.1 SECTION INCLUDES

A. Piping complete with all fittings, joining materials, expansion joints, hangers, supports anchors and necessary appurtenances for interior installations:
   1. Raw water piping
   2. Drain piping

### 1.2 RELATED SECTIONS

A. Section 03300 – Cast-In-Place Concrete

B. Section 03610 – Non-Shrink Grout

C. Section 09900 – Coatings

D. Section 15090 – Supports and Anchors

E. Section 15100 – Valves, Cocks, and Hydrants

### 1.3 REFERENCES

A. American National Standards Institute (ANSI)
   1. B1.1 – Unified Inch Screw Threads
   2. B16.1/B16.5 – Pipe Flanges and Fittings Package
   3. B16.24 – Cast Copper Alloy Pipe Flanges and Flanges Fittings
   4. B16.26 – Cast Copper Alloy Fittings for Flared Copper Tubes
   5. B18.2 – Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series)
   6. B31.1/B31.3 – Power and Process Piping Package

B. American Society of Mechanical Engineers (ASME)
   1. B16.18 – Cast Copper Alloy Solder Joint Pressure Fittings
   2. B16.22 – Wrought Copper and Copper Alloy Solder-Joint Pressure Fittings
   3. B16.26 – Cast Copper Alloy Fittings for Flared Copper Tubes

C. American Society of Testing and Materials (ASTM)
   1. A53/A53M – Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
   2. A134 – Standard Specification for Pipe, Steel, Electric-Fusion (Arc)-Welded (Sizes NPS 16 and Over)
   3. A135 – Standard Specification for Electric-Resistance-Welded Steel Pipe

4. A139 – Standard Specification for Electric-Fusion (Arc)-Welded Steel Pipe (NPS 4 and Over)
5. A181 – Standard Specification for Carbon Steel Forgings for General-Purpose Piping
6. A193 – Standard Specification for Alloy-Steel and Stainless Steel Bolting for High Temperature or High Pressure Service and Other Special Purpose Applications
7. A194 – Standard Specification for Carbon and Alloy Steel Nuts for Bolts for High Pressure or High Temperature Service, or Both
8. A234 – Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service
9. A269 – Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service
10. A283 – Standard Specification for Low and Intermediate Tensile Strength Carbon Steel Plates
11. A307 – Carbon Steel Bolts and Studs, 60,000 psi Tensile Strength
12. A312 – Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes
13. A325 – Standard Specification for Structural Bolts, Steel, Heat Treated, 120/105 ksi Minimum Tensile Strength
14. A774 – Standard Specification for As-Welded Wrought Austenitic Stainless Steel Fitting for General Corrosive Service at Low and Moderate Temperatures
15. A778 – Standard Specification for Welded, Unannealed Austenitic Stainless Steel Tubular Products
16. A795 – Standard Specification for Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless Steel Pipe for Fire Protection Use
17. B32 – Standard Specification for Solder Metal
18. B75 – Standard Specification for Seamless Copper Tube
19. B88 – Standard Specification for Seamless Copper Water Tube
20. D1785 – Standard Specification for Poly Vinyl Chloride (PVC) Plastic Pipe, Schedules 40, 80, and 120
21. D2464 – Standard Specification for Threaded Poly Vinyl Chloride (PVC) Plastic Pipe Fittings, Schedule 80
22. D2466 – Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 40
23. D2467 – Standard Specification for Poly Vinyl Chloride (PVC) Plastic Pipe Fittings, Schedule 80
24. D2996 – Standard Specification for Filament-Wound “Fiberglass” (Glass-Fiber-Reinforced Thermosetting-Resin) Pipe
25. D3035 – Standard Specification for Polyethylene (PE) Plastic Pipe (DR-PR.) Based on Controlled Outside Diameter
26. D3139 – Standard Specification for Joints for Plastic Pressure Pipes Using Flexible Elastomeric Seals
27. D3261 – Standard Specification for Butt Heat Fusion Polyethylene (PE) Plastic Fittings for Polyethylene (PE) Plastic Pipe and Tubing
28. D3350 – Standard Specification for Polyethylene Plastics Pipe and Fitting Materials
29. F104 – Standard Classification System for Nonmetallic Gasket Materials
30. F412 – Standard Terminology Relating to Plastic Piping Systems
31. F477 – Standard Specification for Elastomeric Seals (Gaskets) for Joining Plastic Pipe

32. F714 – Standard Specification for Polyethylene (PE) Plastic Pipe (DR-PR) Based on Outside Diameter
33. F2164 – Standard Practice for Field Leak Testing of Polyethylene (PE) and Crosslinked Polyethylene (PEX) Pressure Piping Systems Using Hydrostatic Pressure

D. American Water Works Association (AWWA)
1. C110 – American National Standard for Ductile-Iron and Gray-Iron Fittings, 3-Inch through 48-Inch, for Water
2. C111 – Standard for Rubber Gasket Joints for Ductile-Iron Pressure Pipe and Fittings
3. C115 – Flanged Ductile-Iron Pipe with Ductile-Iron or Gray-Iron Threaded Flanges
4. C150 – American National Standard for Thickness Design of Ductile-Iron Pipe
5. C151 – Standard for Ductile Iron Pipe, Centrifugally Cast, for Water
6. C153 – Standard for Ductile-Iron Compact Fittings
7. C200 – Standard for Steel Water Pipe 6 Inch and Larger
8. C203 – Standard for Coal-Tar Protective Coatings and Linings for Steel Water Pipelines – Enamel and Tape – Hot Applied
9. C206 – Standard for Field Welding of Steel Water Pipe
10. C207 – Standard for Steel Pipe Flanges for Waterworks Service – Sizes 4 Inch Through 144 Inch
11. C209 – Cold-Applied Tape Coatings for the Exterior of Special Sections, Connections, and Fittings for Steel Water Pipelines
12. C219 – Bolted, Sleeve-Type Couplings for Plain-End Pipe
13. C220 – Standard for Stainless Steel Pipe ½ Inch and Larger
14. C606 – Standard for Grooved and Shouldered Joints
15. C900 – Standard for Polyvinyl Chloride (PVC) Pressure Pipe, 4 Inch through 12 Inch, for Water Transmission and Distribution
16. C901 – Standard for Polyethylene (PE) Pressure Pipe and Tubing, 1/2 In. through 3 In., for Water Service
17. C906 – Polyethylene (PE) Pressure Pipe and Fittings 4 Inch Through 63 Inch, for Water Distribution and Transmission
18. C907 – Injection-Molded Polyvinyl Chloride (PVC) Pressure Fittings, 4 Inch through 12 Inch, for Water, Wastewater, and Reclaimed Water Service
19. C950 – Standard for Fiberglass Pressure Pipe
20. M11 – Steel Pipe: A Guide for Design and Installation
21. M23 – PVC Pipe Design and Installation
22. M41 – Ductile-Iron Pipe and Fittings
23. M45 – Fiberglass Pipe Design

E. National Science Foundation (NSF)
1. 61 – Drinking Water System Components – Health Effects

F. International Plumbing Code (IPC)

G. Copper Development Association Inc. (CDA)
1. Copper Tube Handbook

H. Society for Protective Coatings (SSPC)

## 1.4 SUBMITTALS

A. Shop Drawings:
   1. Provide piping layout fabrication and assembly drawings with fittings dimensions. Provide sufficient information to verify compliance with specification including, but not limited to, the following:
      a. Layout of pipe drawings
      b. Pipe and joint details
      c. Specials, fittings, and coupling details
      d. Specification data sheets
      e. Certificates of compliance with applicable standard and specification and testing certificates
      f. For steel pipe show weld locations and installation of pipe, fittings, specials and connections
         i) Complete data of materials proposed
         ii) Do not manufacture until approved
         iii) Show where each numbered pipe fitting or special is to be installed
         iv) Numbers correspond to those painted on pipe
   2. Provide additional detailed information including elevations, fittings, specialty materials or fabrications for special or custom features, structures, junctions and/or pipes
   3. Specifications, data sheets and affidavits of compliance for coatings and linings
   4. Provide pipe-laying and installation schedule

B. Product Data: Provide sufficient data on features, pipe, joints, gasket material, lubricant and accessories to verify compliance with specifications

C. Manufacturers Certificate and affidavits:
   1. Furnish the following prior to fabrication
      a. Details: Pipe, joint, special, fitting and coupling
      b. Shop coating
      c. Shop coatings and linings: specifications, data sheets, and compliance affidavits
   2. Furnish the following prior to shipment
      a. Compliance affidavit for applicable standards
      b. Test certificates

D. Test Reports: Submit all shop and field test reports to Engineer and Owner for review

## 1.5 PROJECT RECORD DOCUMENTS

A. Accurately record actual locations of piping, valves, connections, centerline of pressure pipe elevations, and top of gravity pipe elevations, and bottom of drain pipe elevations

B. Identify and describe unexpected variations to building conditions or discovery of unknown utilities

## 1.6 REGULATORY REQUIREMENTS

A. In accordance with all local Ouray County codes and ordinances, laws and regulations of the State

B. In case of apparent conflict, State and local requirements govern over these specifications

C. In absence of State and local regulations, International Plumbing Code applies

## 1.7 DELIVERY, STORAGE AND HANDLING

A. For all pipe:
   1. Delivery
      a. Ship rubber gaskets in cartons and store in a clean area away from grease, oil, ozone producing electric motors, heat and the direct sunlight
   2. Storage
      a. Store pipe, fittings and gaskets in clean locations protected from environmental conditions such as: direct sunlight, mud, etc.
      b. Do not use pipe and fittings stored in direct sunlight for periods in excess of 18 months
      c. Store pipe on a flat surface which provides even support for the barrel with bell ends overhanging
      d. Do not stack pipe higher than 5 feet
   3. Storage: Use the following precautions for valves, during storage:
      a. Do not remove end protectors unless necessary for inspection; then reinstall for storage
         i) Protect valves from weather by storing indoors or support valves off ground and pavement in watertight enclosures when outdoor storage is necessary
   4. Handling
      a. Handle so as to insure installation in sound undamaged condition
      b. Use equipment, tools and methods for unloading, reloading, hauling and laying that do not damage pipe or cause an impact. Damaged pipe will be cause for rejection
      c. Use hooks or straps with broad, well padded contact surfaces for lifting sections of pipe
   5. Preparation for Transport:
      a. Prepare valves, for shipping as follows:
         i) Ensure that valves and pilotry are drained and dry and internally protected against rust and corrosion. Protect valves against damage to threaded ends, flange faces, and weld ends. Set valves in best position for handling. Set valves closed to prevent rattling
   6. Deliver and store valves and accessories in shipping containers with labeling in place in accordance with AWWA C500/C504/C509
   7. Seal pipe and fittings to prevent entry of foreign materials into pipe and fittings
   8. During loading, transporting and unloading, exercise care to prevent damage to material
      a. Use nylon slings only
      b. Do not drop pipe or fittings

c. Do not roll or skid against pipe already on ground
d. Repair any damage done to coating or lining
e. Handle per manufacturer's recommendations
f. Store rubber gaskets in cool dark location
g. Store all material on wood pallets or timbers

9. Adequately tag or otherwise mark all piping and fittings as to size

## PART 2 PROCESS PIPING PRODUCTS

### 2.1 PIPE MATERIALS

A. Steel pipe and fittings shall be provided in accordance with ASTM A53/A53M, ASTM A106, or AWWA C200 as specified

### 2.2 DUCTILE IRON PIPE

A. Ductile Iron Process Piping: The following piping shall be ductile iron pipe as indicated on Drawings and provided for locations specified herein:
   1. Irrigation pump suction and discharge piping
      a. Exposed (interior)
   2. Fire suppression system interconnect piping
      a. Exposed (interior)
   3. Irrigation screen filter inlet and outlet piping
      a. Exposed (interior)
   4. Hydropneumatic tank connection
      a. Exposed (interior)

B. Pipe:
   1. AWWA C151, C150, and C151: As listed below except as otherwise specified or indicated on Drawings.
      a. Where fitted with push-on joints, mechanical joints or mechanical joints with joint restraint device, or restrained joints: Class 50
      b. Where fitted with flanged, grooved or restrained joints: Class 53
      c. 4-Inch to 12-Inch: Class 350
      d. Thickness (minimum) for screw-on flanges shall be in accordance with AWWA C115
      e. Pipe with grooved or shouldered joints for any restrained joint shall have wall thickness increased to provide the minimum wall thickness in accordance with AWWA C606

C. Fittings
   1. Ductile iron: ANSI A21.10/AWWA C110, ASTM A536, Grade 80-60-03 or 70-50-05
      a. 24-inch and below: 350 psi rating, mechanical joint or flanged
   2. Ductile iron compact fittings: ANSI A21.53/AWWA C153
      a. 16-inch and below: 350 psi rating, mechanical joint or flanged
   3. Ductile iron grooved fittings: ANSI/AWWA C606, AWWA Fittings, conform to AWWA C110 for center-to-center dimensions and AWWA C153 for wall thickness,

a. 12-inch and below: 350 psi rating, grooved joint

4. Tapping Sleeve and Tapping Valve
   a. Complete assembly, including tapping sleeve, tapping valve, and bolts and nuts. Use sleeve and valve compatible with tapping machine
   b. Tapping Sleeve: Cast-iron or ductile-iron 2-piece bolted sleeve with flanged outlet for new branch connection. Sleeve may have mechanical joint ends with rubber gaskets or sealing rings in sleeve body. Use sleeves that mate with size and type of pipe material being tapped. Outlet flange shall be of a size required for branch connection
5. Provide all specials, taps, flanges, plugs and wall fittings as required
6. Provide openings for air valve, drain, sampling, testing, sensing and all other connections with threaded bosses sized and located as indicated and specified
   a. Threaded connections:
      i) Shall be in accordance with ANSI B2.1.NPT: provide boss or tapping saddle at all tapped connections

D. Joints
1. All joints shall be of restrained type unless otherwise indicated on Drawings
2. Mechanical and push-on joints
   a. Mechanical joints: ANSI A21.11/AWWA C111
      i) Bolts and nuts: High strength, low alloy steel, in accordance with ASTM A307 or accepted substitution
   b. Push-on joints: ANSI A21.11/AWWA C111, except gaskets shall be neoprene or other synthetic rubber. Natural rubber in not acceptable.Gaskets for air piping to be high temperature EPDM (Ethylene Propylenediene Monomer).
      i) Lubricant: Heavy vegetable soap solution suitable for potable water contact
      ii) Pressure rated 350 psi
3. Mechanical joints with tie rods
   a. Tie rods: ASTM A307, galvanized entire length
   b. Steel pipe spacers: ASTM A120, standard weight galvanized
   c. Washers: ANSI A27.2 plain steel, galvanized
4. Flanged joints
   a. Flanged joints shall be provided for all interior and exposed exterior pipe unless otherwise indicated or specified
   b. Flanges
      i) General use: ANSI A21.15/AWWA C115 and ANSI B16.1, Class 125
      ii) Where Class 250 indicated on Drawings or specified: ANSI B16.1, Class 250, flat faced
      iii) Effluent piping with ANSI Class 125 flanges rated for 250 psi
   c. Pipe with Victaulic-style couplings and rigid joints conforming to AWWA C606 may be substituted for Class 125 flanged pipe
   d. Hardware
      i) Hot dipped zinc galvanized steel conforming to ASTM A307, grade B (Class B & D flanges), grade B7 (Class E & F flanges)
      ii) Heads and dimensions per ANSI B1.1
      iii) For mechanical couplings furnish ASTM A183 or A194 bolts and nuts
      iv) Provide two nuts for one-inch diameter and larger bolt applications
5. Mechanical couplings:

    a. Dresser “Style 38”
    b. Rockwell “411”
    c. Or accepted substitution
  6. Flanged coupling adapters 12 inch and under:
    a. Rockwell “Type 912”
    b. Dresser “Style 127”
    c. Or accepted substitution
  7. Tapping saddles:
    a. Ductile iron with double stainless steel straps and rubber sealing gasket, 250 psi pressure rating
  8. Mechanical joint with restraint device
    a. Where approved by the Engineer, multiple wedging action type follower gland used in lieu of tied pipe restraint
    b. Twist off nuts used to insure proper actuating of the restraining device
    c. Working pressure rated to at least 250 psi
    d. EBAA Iron Sales, Inc., MEGA LUG or accepted substitution
  9. Wall castings: Mechanical and/or flanged joints, integral weep ring, sufficient space for bolting of connection piping to joint minimum 6 inches from embedded wall or with tapped holes for follower bolts and plastic plugs to prevent holes from filling with concrete

E. Flanged Coupling Adapters:
  1. Flanged end and body to be one unit conforming to AWWA C219. Coupling end to be compression gland type with follower ring
  2. Adapters for joining exposed interior iron pipe may be steel or iron
  3. Flanged end bolt circle, bolt size, and spacing shall conform to the applicable provisions of ANSI B16.1 and shall be drilled Class 125 for iron adapters. Flanges on steel adapters shall be AWWA C207, Class D, drilled ANSI B16.1 Class 125
  4. Bolts and nuts shall be ductile iron for iron adapters and high-strength, low-alloy steel for steel adapters
  5. Anchor studs shall not be used where joint restraint is required. Furnish adapters with tie rod harness assemblies where indicated
  6. Lining and Exterior Coating for Steel Adapters:
    a. Two-part epoxy or nylon fuse-coated to a minimum 10 mils thickness
    b. Completely coat adapter sleeve and end follower gland plus line interior for adapters intended for exposed exterior, vault or pit installations
    c. Line interior of all adapters intended for exposed interior installations. Coat exterior with normal shop coating

F. Dismantling Couplings/Joints:
  1. Consists of a mechanical joint fitting located between two pipe flanges with restraining rods across the mechanical joint section, providing a restrained system with integral space for removal of adjacent equipment
  2. Shall conform to AWWA C219
  3. Materials shall be steel.
  4. When connected to DIP system, install insulating flange kit

G. Fabrication

1. Joints
    a. Except as indicated on Drawings
        i) Bells in wall pipe or casting: Mechanical with tapped holes for follower bolts or extended minimum 6 inches beyond wall
        ii) Exposed: Flanged or grooved
    b. 12 inch and smaller branch outlets where main line is at least twice the diameter of the branch, tee or a tapping saddle is acceptable
    c. Tie rods may be required as indicated on Drawings
    d. Mark the centerline of each flange and mechanical joint piece
    e. Screw flanges onto screwed-on flanged pipe so that pipe extends completely through and flush with the flange
    f. Finish machine pipe ends and flange faces flat and perpendicular to pipe centerline in a single operation
2. Shop coating and lining
    a. Contractor option to shop prime and coat the interior surfaces of all process piping 16 mil thickness with high build coal tar epoxy except where used for gas and air service
    b. Cement line the interior surfaces of fittings, except for air piping
    c. Shop prime exterior surfaces of pipe and fittings to be installed in exposed interior or exterior locations
    d. Shop coat flange faces with rust preventive compound

## 2.3 HIGH DENSITY POLYETHYLENE (HDPE) PRESSURE PIPING

A. HDPE pipe shall be as indicated on the drawings and specified herein:
    1. As indicated in Piping Schedule below

B. Manufacturers:
    1. Performance Pipe
    2. WL Plastics
    3. ISCO Pipe
    4. Or accepted substitution

C. Material
    1. The pipe shall be manufactured from a PE 4710 resin compound listed with the Plastic Pipe Institute (PPI) as TR-4.
    2. The resin material shall be in accordance with ASTM D3350 with a minimum cell classification of 445574C/E.
    3. This resin material shall have a Long Term Hydrostatic Strength of 1600 PSI when tested in accordance to ASTM D2837.
    4. Pipe dimensions shall be in accordance with ASTM D3035 as a minimum.
    5. The final compounded material shall contain a minimum of 2% carbon black
    6. The pipe shall contain no recycled material except that generated by the pipe manufacturer in their own plant from resin compound of the same specification and raw material supplier. The pipe shall be homogeneous throughout and free of visible cracks, holes, foreign inclusions, voids, or other injurious defects.
    7. Pipe shall have a manufacturing standard of ASTM F714 and be manufactured by an ISO 9001 certified manufacturer.

8. All pipes and fittings shall be suitable for use as pressure conduits, listed as NSF 14, and per AWWA C906 Pressure Class (PC) 100 have a nominal burst value of three and one-half times the Working Pressure Rating (WPR) of the pipe and/or fitting
9. The pipe shall be DR11 unless noted otherwise on the Drawings

D. Fittings
1. All fittings shall be PE 4710 HDPE, minimum Cell Classification of 445574C/E as determined by ASTM D3350, and approved for AWWA use.
2. All fittings shall be of the same base resin as the pipe.
3. All fittings shall have a working pressure rating equal to the pipe unless otherwise specified in the plans.
4. All fittings shall be homogeneous throughout and free of visible cracks, holes, foreign inclusions, voids, or other injurious defects.
5. Butt Fusion Fittings
   a. Molded butt fusion fittings shall be in accordance with ASTM D3261 and shall be manufactured by injection molding, a combination of extrusion and machining, or fabricated from HDPE pipe conforming to this specification.
6. Compression Type Mechanical Coupling
   a. Suitable for joining HDPE to HDPE, HDPE to PVC, or HDPE to DIP
   b. Factory coat coupling internally and externally with a fusion bonded epoxy
   c. Reinforce HDPE pipe with a split ring type stiffener in pipe bore
      i) Size stiffeners for size of HDPE pipe being joined
      ii) Supply feature that prevents stiffener from sliding completely into pipe
      iii) Size stiffeners for length of mechanical coupling and not to extend outside of body of mechanical coupling
      iv) Mark stiffener with pipe diameter
      v) Factory coat stiffeners internally and externally with fusion bonded epoxy
   d. Use seal and restraint type coupling. Requirements for type of couplings are specified herein or shown on Drawings
      i) Approved Manufacturers:
         a) JCM Industries
         b) Agru America
         c) Romac Industries
         d) Or accepted substitution

E. Transition Couplings: One piece assembly
1. One end being HDPE pipe with butt fusion joining technique
2. Other end being either steel or brass pipe threaded suitable for connecting ductile iron, threaded fittings, or threaded valves
   a. Approved Manufacturers:
      i) Central Plastics Company
      ii) Industrial Pipe Fittings, Inc.
      iii) Or accepted substitution

F. Pipe fittings and flanged connections, to be joined by thermal butt-fusion, shall be of the same type, grade, and class of polyethylene compound and supplied from the same raw material supplier.

G. Sidewall fusions for connections to outlet piping shall be performed in accordance with HDPE pipe and fitting manufacturer's specifications. The heating irons used for sidewall fusion shall have an inside diameter equal to the outside diameter of the HDPE pipe being fused. The size of the heating iron shall be ¼ inch larger than the size of the outlet branch being fused.

H. Socket fusion, hot gas fusion, threading, solvents, and epoxies will not be used to join HDPE pipe

I. Flange Gaskets
   1. Flange gaskets shall be full-faced and shall be in accordance with ASTM D1330
   2. Flange gasket drilling pattern shall conform to ANSI B16.1/B16.5
   3. Flange gaskets shall be 1/16" thick for flanges up to 14" diameter. 1/8" thick gaskets shall be provided for flanges over 14" diameter
   4. Gasket material shall be EPDM

2.4 ACCESSORIES

A. Watertight and dust-tight pipe sleeves:
   1. O-Z Electrical Manufacturing Company
   2. Thruwall and Floor Seals
   3. Or accepted substitution

B. Modular, rubber, sealing elements:
   1. Thunder Line Corporation "Link Seal"
   2. Or accepted substitution

C. Thread tape: Teflon
   1. John Crane "Thread Tape"
   2. Garlock "Plasti-Thread"
   3. Hoke "EZ Seal"
   4. Or accepted substitution

D. Sealant: Silicone sealant as specified in caulking section

E. Pulsation Snubbers:
   1. Operating and Maintenance Specialties "Ray Pressure Snubbers"
   2. Ashcroft "Snubber Series 25-255"
   3. Or accepted substitution

F. Gauge cocks: Bronze, tee handle
   1. Lunkenheimer 1178
   2. Powell 915
   3. Or accepted substitution

G. Concrete for supports and other locations as shown on Drawings: Concrete type specified in Section 03300

## PART 3 EXECUTION

### 3.1 INSTALLATION

A. Install as specified and indicated on Drawings

B. Remove dirt, rocks, debris, and other foreign materials from all pipelines

C. Keep interior of pipe and fittings thoroughly clean before installation and until work is accepted

D. Take precautions to prevent entrance of foreign material during jointing, lining repair, and inspection operation

E. Seal open end of line with watertight plug if pipe laying stopped

F. Provide a shutoff valve and union at the water and air supply connections to each fixture and unit of equipment, whether shown on the Drawings or not

G. Provide a union within 2 feet of each threaded end valve unless other means for easy removal of the valve are available

H. Provide unions where required for equipment removal, indicated on the Drawing or specified

I. Do not install piping to obstruct openings and passageways

J. Cut pipe to measurement taken at the site, not from the Drawings

K. Layout piping to provide for expansion and contraction

L. Provide taps for pressure gauge connections with a nipple, snubber, and gauge cock

M. Securely anchor piping at the midpoints between expansion joints

N. Provide air supply piping with sectionalizing valves and valved air inlet connections to isolate portions of system for periodic testing

O. Grade all air piping to points of drainage; provide driplegs on drain valves at drainage points

P. Uniformly grade all piping serving metering equipment and provide complete venting to eliminate air traps

Q. Pipe stuffing box leakage from water sealed pumps to nearest point of drainage

R. Provide insulating fittings and dielectric isolation in all piping except air and gas piping wherever copper tubing or fittings are connected to iron or steel pipe or fittings

S. Provide expansion joints where restrained or flanged pipes or fittings prevent normal joint movement

## 3.2 CONNECTION TO EXISTING PIPELINES

A. Make connections between new and existing piping with suitable fittings

B. Schedule connection to minimize inconvenience to the Owner

C. Undertake connections in order to disturb the system as little as possible

D. Where connection involves potable water systems, provide disinfection methods as prescribed in these Specifications

E. Once tie in to existing system is initiated, continue Work continuously until tie-in is made and tested

## 3.3 JOINTS

A. Make pipe joints carefully and neatly

B. Connect piping in accordance with manufacturer recommendations

C. Threaded
   1. ANSI B2.1, NPT fully and cleanly cut with sharp dies
   2. No more than 3 threads exposed after installation
   3. Ream pipe ends after threading to remove burrs
   4. Apply Teflon thread tape to joints in all plastic and stainless steel piping
   5. Apply thread tape or joint compound to joints in other piping
   6. Apply Teflon thread tape or litharage and glycerine paste to joints on steel piping for chlorine service

D. Compression
   1. Cut pipe ends squarely, remove burrs
   2. Clean contact surfaces with steel wool

E. Flared
   1. Cut tubing ends squarely, remove burrs
   2. Scratches or grooves in flared ends are not allowed

F. Soldered and brazed
   1. Braze joints in 2 inch or larger copper tubing
   2. Solder or braze lines smaller than 2 inch where solder fittings are specified
   3. Thoroughly clean joint surfaces with flint paper and coat with thin film of flux
   4. Install tubing to full depth of socket
   5. Do not overheat metal or flux
   6. Uniformly heat joint to melt filler metal on contact
   7. Remove surplus filler metal and flux while joint is still hot

G. Solvent welded
   1. Cut PVC pipe ends square and smooth, wipe clean
   2. Remove all burrs and filings
   3. Using a small brush, apply cement primer to the outside of pipe and the inside of the fitting socket
   4. Using a small brush, apply solvent cement to outside of pipe and inside of the fitting socket
      a. Solvent cement to be specifically rated for use with chlorine or hypochlorite solutions for piping and fittings that will be used to convey these solutions
   5. To ensure uniform cement distribution, push the coated surface snugly together and rotate approximately 1/2 turn
   6. Remove excess cement by wiping

H. Grooved couplings
   1. Cut grooves with grooving tool
   2. Groove to rigid grooving dimensions
   3. Groove cleanly and sharply without burrs or check marks
   4. Form rounded bottom "radius grooves" in plastic piping

I. Welded: ANSI B31.1, and per applicable Standards for pressure piping

J. Push on
   1. Per recommendations of pipe manufacturer
   2. Bevel each spigot end to facilitate assembly
   3. Lubricate with a heavy vegetable soap solution immediately before joint is completed
   4. Store lubricant in closed containers, keep clean, suitable for use in potable water contact

K. Mechanical Joints
   1. If an effective seal is not obtained, disassemble joint, clean thoroughly, and reassemble
   2. Do not over tighten bolts to compensate for poor installation
   3. Carefully align holes in mechanical joints with restraint device to permit installation of the harness bolts
   4. Install flange and mechanical joint pieces so the four mechanical joint holes, as well as the flange holes, straddle the top centerline for horizontal piping, or the side centerline for vertical piping

L. Flanged joints
   1. Take care when bolting flanges to insure that there is no restraint on the opposite end of the pipe which would prevent gasket compression or cause unnecessary stress in flanges
   2. Leave one flange free to move in any direction while tightening flange bolts
   3. Do not pack or assemble bell and spigot joints until all flanges affected thereby have been tightened
   4. Tighten bolts gradually at a uniform rate to compress gaskets uniformly

## 3.4 PIPE SLEEVES

A. Provide for pipes passing through concrete or masonry

B. Install before concrete is placed

C. Through ceramic or vinyl-tile floors, install sleeves flush with finished floor and provide nickel or chromium plated floor plates

D. Where passing through all other floors, install so the sleeve projects between 1 and 2 inches above the floor

E. Seal sleeves passing through slabs with 1 side against soil with a modular sealing element or a watertight pipe sleeve

F. If insulated, extend insulation through sleeves

G. For future pipe installation, provide sleeves and seal ends with plastic caps or plugs

H. For piping through interior walls and floors with special finish provide pipe sleeves or holes drilled with rotary drill

I. Make dust and gas tight through room walls and floor

J. Six inch or smaller; special dust-tight sleeves

K. Greater than 6 inches, seal with modular sealing elements or caulk with backer rod and seal both sides with non-sag silicone sealant
   1. For pipe sleeves located in water bearing walls provide wall pipes or sleeves with dual modular sealing elements as shown on the Drawings

3.5 PROTECTIVE COATING

A. Paint exposed, threaded, or submerged galvanized steel piping with zinc-rich paint that have cut ends or are nicked or gouged

3.6 CLEANING

A. General
   1. The inside of all pipe, valves, and fittings shall be smooth, clean, and free from blisters, loose mill scale, sand, and dirt when erected
   2. Immediately prior to pressure testing, clean and remove grease, metal cuttings, dirt, or other foreign materials which may have entered the system
   3. Flush or blow all lines thoroughly before placing in service
   4. At completion of Work and prior to final acceptance, thoroughly clean Work installed under these specifications
      a. Clean equipment, fixtures, pipe, valves, and fittings of grease, metal cuttings, and sand and dirt which may have accumulated by operation of system, from testing, of from other causes
      b. Repair any stoppage or discoloration or other damage to parts of building, its finish, or furnishings, due to failure to properly clean piping system, without cost to Owner

## 3.7 FIELD QUALITY CONTROL

A. General
   1. Utilize pressures, media, and pressure test durations as specified on Piping Schedules
   2. Isolate equipment which may be damaged by the specified pressure test conditions
   3. Perform pressure test using calibrated pressure gauges and calibrated volumetric measuring equipment to determine leakage rates. Select each gauge so that the specified test pressure falls within the upper half of the gauge range. Notify Engineer 24 hours prior to each test
   4. Completely assemble and test new piping systems prior to connection to existing pipe systems
   5. Acknowledge satisfactory performance of tests and inspections in writing to Engineer prior to final acceptance
   6. Provide all necessary equipment, materials, tools, appliances, devices, and perform all work required in connection with the tests and inspections
   7. Bear the cost of all testing and inspecting, locating and remedying of leaks and any necessary retesting and re-examination

B. Pipe leaks
   1. All joints and seams whether tested or not shall be watertight and airtight
   2. Inspect all exposed shop and field welded seams
   3. Leaks shall be clearly marked
   4. Welded joints (steel) shall be repaired by chipping out defective parts and rewelding. Welds shall not be hammered
   5. Leaks discovered by Owner within one (1) year of final acceptance shall be repaired by the Contractor at no additional expense to the Owner

C. Test each line at the Contractor's expense in the presence and to the satisfaction of Engineer

D. General testing methods and criteria
   1. Types of pressure testing and inspection to be employed include hydrostatic pressure testing, cylinder water pumped compressed air or cylinder nitrogen testing, low pressure air testing and hydrostatic exfiltration/infiltration testing
   2. Liquid systems
      a. The following liquid piping systems shall have zero allowable leakage at the specified test pressure throughout the specified duration of 2 hours: exposed piping and buried or exposed piping carrying liquid chemicals or water
      b. Hydrostatic pressure testing
         i) For exposed piping: Prepare each section for testing, using adequate bracing and protect system equipment susceptible to damage by test pressures. Do not paint or insulate exposed piping until successful performance of pressure test.

E. Dielectric Testing Methods and Criteria
   1. Provide electrical check between metallic non-ferrous pipe or appurtenances and ferrous elements of construction to assure discontinuity has been maintained
   2. Wherever electrical contact is demonstrated by such test, locate the point or points of continuity and correct the condition

3. Check the integrity of each cad-welding connection using a light hammer blow at a 45 degree angle

## 3.8 PIPING SCHEDULE

Install interior pipes as scheduled in the pipe schedule provided below

| Description | Size (inches) | Material | Acceptable Fitting Type |
|---|---|---|---|
| Irrigation Pump Suction Pipe | 6 | DIP | Flanged, MJ |
| Irrigation Pump Discharge Pipe | 4 | DIP | Flanged, MJ |
| Fire Suppression System Interconnect Pipe | 4 | DIP | Flanged, MJ |
| Screen Filter Discharge Pipe | 4 | DIP | Flanged, MJ |
| Hydropneumatic Tank Connection | 2 | DIP | Flanged, MJ |
| Intake Screen Backwash Pipe | 1 | HDPE | Compression/Fused |
| Screen Filter Backwash Pipe | 2 | HDPE | Fused |
| Sump Pipe | 2 | HDPE | Fused |

END OF SECTION

SECTION 15100

VALVES, COCKS, AND HYDRANTS

## PART 1 - GENERAL

### 1.1 SECTION INCLUDES

A. Valves indicated on the drawings, specified, or required for proper operation of equipment or systems

### 1.2 RELATED SECTIONS

B. Section 09900 – Coatings

C. Section 15060 – Pipe and Pipe Fittings

D. Section 16050 – Basic Electrical Materials and Methods

### 1.3 REFERENCES

A. American National Standards Institute (ANSI):
   1. B16.1 – Cast Iron Pipe Flanges and Flanged Fittings

B. American Society for Testing and Materials (ASTM):
   2. A48 – Standard Specification for Gray Iron Castings
   3. A126 – Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings
   4. A216 – Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service
   5. A536 – Standard Specification for Ductile Iron Castings
   6. B62 – Standard Specification for Composition Bronze or Ounce Metal Castings

C. American Water Works Association (AWWA):
   7. C111 – Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings
   8. C207 – Steel Pipe Flanges for Waterworks Service, Sizes 4 In. Through 144 In.
   9. C500 – Metal-Seated Gate Valves for Water Supply Service
   10. C502 – Dry-Barrel Fire Hydrants
   11. C503 – Wet-Barrel Fire Hydrants
   12. C504 – Rubber Seated Butterfly Valves
   13. C508 – Swing-Check Valves for Waterworks Service, 2 In. Through 24 In. NPS
   14. C509 – Resilient-Seated Gate Valves for Water Supply Service
   15. C510 – Double Check Valve Backflow Prevention Assembly
   16. C511 – Reduced-Pressure Principal Backflow Prevention Assembly
   17. C512 – Air Release, Air/Vacuum, and Combination Air Valves for Waterworks Service
   18. C514 – Air Valve and Vent Inflow Preventer Assemblies for Potable Water Distribution System and Storage Facilities

19. C515 – Reduced-Wall, Resilient-Seated Gate Valves for Water Supply Service
20. C517 – Resilient-Seated Cast-Iron Eccentric Plug Valves
21. C518 – Dual-Disc Swing-Check Valves for Waterworks Service
22. C520 – Knife Gate Valves, Sizes 2 In. (50 mm) Through 96 In. (2,400 mm)
23. C530 – Pilot-Operated Control Valves
24. C541 – Hydraulic and Pneumatic Cylinder and Vane-Type Actuators for Valves and Slide Gates
25. C542 – Electric Motor Actuators for Valves and Slide Gates
26. C550 – Protective Interior Coatings for Valves and Hydrants
27. C560 – Cast-Iron Slide Gates
28. C561 – Fabricated Stainless-Steel Slide Gates
29. C562 – Fabricated Aluminum Slide Gates
30. C563 – Fabricated Composite Slide Gates

1.4 SUBMITTALS

A. Product Data: Provide data on valves and accessories. Provide manufacturer's catalog information with dimensions, materials, and assembled weight. Indicate valve data and pressure ratings

B. Manufacturer's Instructions: Provide complete manufacturer's installation instruction

1.5 PROJECT RECORD DOCUMENTS

A. Record actual size, type and locations of valves

1.6 OPERATION AND MAINTENANCE DATA

A. Maintenance Data: Include installation instructions, spare parts list, exploded assembly views, maintenance requirements

1.7 QUALITY ASSURANCE

A. Manufacturer's name and pressure rating marked on valve body

1.8 QUALIFICATIONS

A. Manufacturer: Company specializing in manufacturing the products specified in this section with minimum 10 years documented experience

1.9 REGULATORY REQUIREMENTS

A. Conform to all municipal codes and ordinances, laws and regulations of the State

B. In case of apparent conflict, State and local requirements govern over these specifications

C. In absence of State and local regulations, International Plumbing Code will apply

1.10 DELIVERY, STORAGE, AND HANDLING

A. Prepare valves and accessories for shipment according to AWWA C500

B. Accept valves on site in shipping containers with labeling in place. Inspect for damage

C. Provide temporary protective coating on cast iron and steel valves

D. Seal valve ends to prevent entry of foreign matter into valve body

E. Drain valves and pilotry of water prior to shipping and storage

PART 2 PRODUCTS

2.1 GENERAL

A. Construction
   1. Actual valve length within ± 1/16" of specified or theoretical length
   2. Ends, except as otherwise specified
      a. 3" and smaller, threaded or soldered ends
      b. 4" and larger, flanges, ANSI 125 lb
      c. Wafer style valves designed for installation between ANSI 125 lb flanges

B. Actuators
   1. Manual actuators:
      a. Provide manual actuators for all valves not specified to be power actuated or designed for automatic operation
         1) Plug Valves: Handwheel
         2) Butterfly Valves: Quarter-turn lever
         3) Seven feet or more above the floor or grade unless otherwise indicated: Chain wheel with operating chain
            a) Equipped with chain guide to permit rapid chain handling of the chain without “gagging” the wheel and to permit reasonable side pull on the chain
            b) With extensions as required to prevent interference with adjacent piping or equipment
            c) Chain heavily zinc or cadmium-plated and looped to extend between 3' and 4' off the floor or grade below valve
      b. Buried valves specified in this section
      c. Rotation
         1) Counter clockwise (to the left) to open
         2) The word “OPEN” and an arrow indicating the direction to open cast on each valve body or operator
         3) Position indicator clearly visible from operating floor
      d. Extension stems
         1) Provide where indicated on drawings, specified, or required for proper operation

2) Non-rising stems
   a) Solid steel shafting with O.D. not less than O.D. of valve stem or galvanized steel pipe with I.D. not less than O.D. of valve stem
   b) Connected to the valve by a flexible socket coupling
   c) All other connections pinned, keyed, or socket
3) Rising stem
   a) Stainless steel or carbon steel shafting with O.D. not less than O.D. of valve stem
   b) Bronze or stainless steel sleeves securely attached to stem
   c) Sleeve length and location to extend through each stem guide throughout the full vertical travel of the stem

e. Stem guides
   1) Cast iron, bronze brushed, adjustable in two directions
   2) If extension stem length exceeds 10' or the weight exceeds 50 lbs., design top guide to carry the stem weight and provide a collar on the stem to bear against the thrust guide
   3) Max spacing
      a) Non-rising stems: 100 times stem O.D.
      b) Rising stems: 60 times stem O.D.
      c) Ten-foot maximum

f. Operating Stands
   1) Provide as indicated on Drawings
   2) Fabricated steel or cast iron
   3) Support the handwheel or lever approximately 36" above the floor
   4) Handwheel diameter: 8" minimum
   5) Provide a standard weight galvanized pipe sleeve for opening in the floor
   6) For longer than 10' and without stem guides with thrust bearings, provide a suitable thrust bearing in each stand to carry extension stem weight

## 2.2 ECCENTRIC PLUG VALVES

A. Manufacturers
   1. Valves
      a. DeZurik
      b. Flowserve
      c. Henry Pratt Company
      d. Milliken
      e. Val-Matic Valve and Manufacturing Corporation
      f. Or Engineer accepted substitution

B. Valve Design
   1. Valves shall conform to design specified herein and in Valve Schedule provided in this section
   2. Quarter-turn non-lubricated eccentric plug valves
   3. Resilient faced plug
   4. Tight shut-off up to scheduled rating with pressure in reverse direction where scheduled
   5. Valves with vane type seat rings are not acceptable

6. Valve port shall be at least 80% of full pipe area for valves through 20-inch in diameter
7. Valve ends to match connecting piping
   a. Buried: Mechanical joint, ANSI A21.11/AWWA C111
   b. Flanged: 125 lb, ANSI B16.1
   c. Screwed valve ends shall be to the NPT standard
   d. Grooved ends shall conform to AWWA C606 rigid joint specifications
8. Minimum Working Pressure Rating:
   a. 175 psig, 4 inch through 12 inch
9. Opening motion eccentric, lifting plug away from body seat
10. Valve Alignment
    a. In horizontal pipelines, valve shall be installed so that the plug is horizontal and rotates upward as the valve opens
    b. In vertical pipelines, valve shall be installed with the end marked "seat" at the top of valve
    c. Valve shall be installed with seat on low pressure side of valve
11. Provided with fully adjustable plug position stops
12. Plugs shall be eccentric type with no backing ring or frame
13. Valve body cavity shall be smooth without protrusions or baffles
14. Electric actuators shall conform to AWWA C542
15. Valve bonnet stuffing box sufficiently deep for 4 packing rings on 4" and larger valves
16. Valve body plainly marked to indicate seat end
17. Actual length within 1/16" ± of specified or theoretical length
18. Valve packing adjustment accessible without removing actuator from valve

C. Valve Materials
1. Plug and body: Cast iron, ASTM A126, Class B, or Ductile iron, ASTM A536, Grade 65-45-12
2. Resilient plug facing or replaceable style body seats shall be synthetic rubber, neoprene, or Buna N compound suitable for use with water and wastewater applications
3. Seat rings shall be threaded, or welded of corrosion-resistant stainless steel (18-8), nickel, or Monel conforming to AWWA C517
4. Sprayed or plated mating seat surfaces are not acceptable
5. Bearings shall be replaceable. Sleeve type and thrust bearings in the upper and lower journals shall be corrosion-resistant stainless steel or bronze
6. Shaft seals shall be multiple O-ring, self-adjusting U-cup or chevron type packing conforming to AWWA C517
7. Pull-down packing is not acceptable
8. Shaft seals shall be field adjustable or replaceable without valve disassembly
9. Plug seat: Chloroprene (Neoprene)
10. Packing: Acrylonitrile Butadiene V-Type Cup
    a. Dual U-cup
11. Upper thrust bearing: TFE
12. Body seat: Welded-in overlay seat of no less than 90% nickel
13. Upper and lower trunnion bearings: Sleeve type, 18-8 stainless steel

14. Valves complete with epoxy coating on the interior and exterior

D. Testing
1. Each valve shall be in accordance with AWWA C517 subjected to cycle life and pressure leak test (1,034 kPa)(150 psi) and a valve body hydrostatic test (2,068 kPa)(300 psi) by the manufacturer at their facilities prior to shipping
2. Valves shall be capable of drip-tight shut-off up to the full leak test rating
   a. Test and certify pressure capacity in the reverse direction

E. Actuators
1. Manual Actuators
   a. Provide manual actuators for all valves not specified to be power actuated or designed for automatic operation
      1) Actuators shall be totally-enclosed oil or grease-lubricated worm gear or traveling nut type
         a) Worm gear actuators
            i) Shall be totally enclosed, grease sealed, gear type
            ii) Shall be furnished with AWWA nut, crank, handwheel, or chainwheel
            iii) Shall be sized to meet the torque ratings of AWWA C504
      2) Actuators shall have AWWA input stops. Actuator shall be self-locking at all variable opening positions
      3) General use: Handwheel
      4) Valve box and nut operator
         a) Buried vales shall be operated through valve boxes and as indicated on drawings: Wrench nuts
         b) Non-rising stems:
            i) Solid steel shafting with O.D. not less than O.D. of valve stem or galvanized steel pipe with I.D. not less than O.D. of valve stem
            ii) Connected to the valve by a flexible socket coupling
            iii) All other connections pinned, keyed, or socket
         c) Stem guides
            i) Cast iron, bronze brushed, adjustable in two (2) directions
            ii) If extension stem length exceeds 10' or the weight exceeds 50 lbs., design top guide to carry the stem weight and provide a collar on the stem to bear against the thrust guide
         d) Max-spacing: Non-rising stems: 100 times stem O.D.
         e) Provide spaces to center stem in valve box
         f) Provide 2 operating keys and wrench nut
         g) Valve boxes
            i) Boxes shall consist of a cast iron cover, lid, and base castings. No slip type boxes shall be allowed
            ii) Type: Cast iron or ductile iron, extension sleeve (screw) type
            iii) Minimum thickness: 3/16" at any point
            iv) Coating: Bituminous varnish
            v) Cast appropriate name designation of service in cover.
            vi) Shaft shall be ductile iron and minimum diameter of 5-¼"
            vii) An appropriate word designating the valve service case on the cover

h) Valve box manufacturers
   i) Neenah Foundry Company
   ii) Tyler Company
   iii) Mueller Company
   iv) Or Engineer accepted substitution

b. Rotation
   1) Valves shall open counterclockwise
   2) The word “OPEN” and a raised cast arrow indicating the direction to open cast on each valve body or operator
   3) Position indicator clearly visible from operating floor

c. Actuators and handwheels shall be located in positions indicated or as otherwise determined when manufacturer's drawings are submitted to Engineer

## 2.3 BUTTERFLY VALVES (WATER AND PROCESS SERVICE)

A. Manufacturers:
   1. DeZurik
   2. Milliken
   3. Val-Matic
   4. Or Engineer accepted substitution

B. Valve Design and Conditions
   1. Bi-directional pressure rating: 150 psi for 3 inch through 24 inch valves
   2. Temperature rating: 250°F
   3. Bi-directional and tested to 110% of full rating
   4. Shall be in conformance with AWWA C504 and C542
   5. Conform to Valve Schedule in this section
   6. Valve ends:
      a. Shall be ANSI B16.1 Class 125 for cast-iron body valves Class 25A through Class 150B
      b. Shall be ANSI B16.1 Class 125 for ductile iron body Class 250 valves
      c. Shall be AWWA C207 Class D with ANSI B16.1 Class 125 drilling for steel body Class 25A through Class 150B valves
      d. Shall be AWWA C207 Class E for steel body Class 250 valves
   7. Joints:
      a. Shall be mechanical or push-on type rubber-gasket joint ends
      b. Shall conform to AWWA C111
   8. The use of a stop or lug cast integrally with or mechanically secured to the body for the purpose of limiting disc travel by means of direct contact or interference with the valve disc (in either the open or closed position) will not be acceptable

C. Materials
   1. Body:
      a. Cast iron for Class 25A through Class 150B valves
      b. Ductile iron for Class 250
   2. Disc:
      a. Cast iron or ductile iron for Class 25A through Class 150B valves
      b. Ductile iron for Class 250

3. Seats:
    a. NBR for services up to 180 Deg F
    b. Shall be body mounted
    c. Shall be resilient type
    d. Mating seat surface shall be stainless steel or Monel
        1) Sprayed or plated mating surfaces are not acceptable
4. Shaft: 316 stainless steel
5. Shaft seals:
    a. Shall be designed for use of standard self-adjusting split-V-type packing or Viton O-ring seals
    b. Pull down packing is not acceptable

D. Valve position indicators
1. Provide position indicators on all exposed operators
2. Provide handwheel with an arrow with the word open on handwheel rim specifying opening direction
    a. Arrow shall be cast on rim and raised
3. Identify open position of valve, 0 - 100%

E. Operator locking devices
1. Modulating service: Infinitely variable locking device or a totally enclosed geared operator as indicated

## 2.4 GATE VALVES – 3" TO 12" AND ACCESSORIES

A. Manufacturers:
1. Mueller
2. American AVK
3. American Flow Control
4. Clow
5. Kennedy
6. Or accepted substitution

B. AWWA C509, Iron body, bronze trim, two O-ring stem seals, non-rising stem with square nut, single wedge, resilient seat, mechanical joint ends, extension stem, and extension valve box, pressure rating of 250 psi. For installation in horizontal or near horizontal pipe lines
1. Non-adjustable elastomeric stem seals
    a. Adjustable packing glands are not permitted
2. Direct operation of stem from above via 2-inch square nut
    a. No gear box provided
3. Rotation: Counterclockwise to open with the word "OPEN" and an arrow indicating the direction to open cast on valve body or operating nut

C. Rotation: Contractor to verify the operating direction with the owner
1. Provide the word "OPEN" and an arrow indicating the direction to open cast on valve body or operating nut.
2. Contractor to confirm nut size with the owner.

D. Valve stem material: ASTM B763, UNS alloy C99500 minimum yield strength of 40,000 psi
   1. Valve stem extensions: Provide valve stem extensions as necessary for proper valve operation with a 7-foot key with tee handle
   2. Provide one (1) key to Owner prior to project closeout

E. Extension stems
   1. Provide for buried valves with operating nuts more than 4.5 feet below grade
   2. Non-rising stems
      a. Solid steel shafting with O.D. not less than O.D. of valve stem or galvanized steel pipe with I.D. not less than O.D. of valve stem
      b. Connected to the valve by a flexible socket coupling
      c. All other connections pinned
      d. Extend stem to within 6-inch of grade
      e. Provide spacers to center stem in valve box
      f. Provide wrench nut

F. Coating
   1. AWWA C550 and NSF 61 certified
   2. Minimum 8 mils dry film thickness
   3. Fusion bonded epoxy applied to all ferrous metal surfaces after cleaning surfaces of grease, dirt and moisture, and performing near-white blast cleaning following SSPC-SP10
   4. Do not coat fasteners or machined surfaces subject to contact and relative movement against other surfaces during operation of valve or other surfaces where such coating would compromise proper installation or functionality of valve

G. Valve boxes, depth as required for valve
   1. Three piece cast iron (complying with ASTM A48, Class 20A) adjustable screw type, 5.25-inch diameter, minimum thickness of 3/16 inch
   2. Box, cover, and base coated by dipping in asphalt varnish.
   3. Cover marked with word, “WATER.”
   4. Provide extension piece to permit 6-inch adjustment above finish grade
   5. Manufacturers:
      a. Tyler Pipe Company "Series 6860 with #160 oval base"
      b. East Jordan Iron Works "8560 Series"
      c. Tyler Union "6860 Series"
      d. Or accepted substitution

## 2.5 BALL CHECK VALVE

A. Manufacturers
   1. Flomatic Valves, Type 408
   2. Flygt Corporation, Type 5087
   3. AVK, Series 53
   4. Or Engineer accepted substitution

B. Design

1. Ball check valves shall be designed to be non-clog, fully automatic, maintenance free, and specifically suited for operation in wastewater where solids, fibers, grit or highly viscous materials are encountered
2. Ball check valves are designed to be maintenance-free and suited for installation in the horizontal or vertical position
3. Ball check valves will have one moving part, the ball, which automatically rolls out of the path of flow, thus providing an unobstructed and "full flow" equal to nominal size. Upon discontinuation of flow, the ball shall automatically roll back to the closed position providing a positive seal against back pressure or backflow
4. Ball shall have an exterior coating of vulcanized nitrile rubber resistant to grease, petroleum products, animal and vegetable fats, dilute concentrations of acids and alkalines, tearing and abrasion
5. Valve shall be constructed so that by unbolting and lifting off the cover, the ball may be removed and replaced without removing the valve from the line
6. Ball check valves will be available with either a floating or sinking ball.
7. Suitable for 150 psi working pressure (all sizes)
8. Conform to Valve Schedule criteria in this section
9. Body and Cover: Flanged iron body shall be nodular cast iron type
10. Flanges: Shall be ANSI B16.1 Class 125

## 2.6 SWING CHECK VALVES

A. Manufacturers:
   1. Val-Matic Swing-Flex
   2. Or Engineer accepted substitution

B. Design
   1. 250 psi working pressure
   2. Conform to AWWA C508
   3. Conform to Valve Schedule criteria in this section
   4. Body and Cover: Shall be ductile iron, bronze mounted, full opening
   5. Flanges: Shall be ANSI B16.1 Class 125
   6. Furnish with mechanical indicator to provide disc position on valves 3” and larger. Indicator shall have continuous contact with the disc under all operating conditions to assure accurate disc position indication.

## 2.7 BALL VALVES (BRONZE OR STAINLESS STEEL)

A. Manufacturers for air and water service 2-inches and smaller:
   1. Crane
   2. Jamesbury (Metso)
   3. Jenkins (Crane)
   4. Nibco
   5. Or Engineer accepted substitution

B. Design:
   1. 175 psi working pressure (general)
   2. Conform to Valve Schedule criteria in this section

C. Application
   1. Except as otherwise indicated or specified, provide ball valves for all 2-1/2" or smaller shutoff valves
   2. Provide additional ball valves as indicated on Drawings or as specified

D. Fabrication
   1. Handles showing direction of opening
   2. Ball design which does not allow media contact with stem
   3. Balancing stop for all applications

## 2.8 COMBINATION AIR RELEASE AND VACUUM RELIEF VALVES

A. Sump Pump Discharge
   1. Provide as indicated on Drawings
   2. Water working pressure: 125 psi
   3. Manufacturer
      a. A.R.I.
      b. Plast-o-Matic
      c. Hayward
      d. Or Engineer accepted substitution
   4. Provide a shutoff valve: Ball valve, gate valve, butterfly valve, eccentric action handle
      a. Manufacturers:
         1) Nibco
         2) Dyna-Quip
         3) Or Engineer accepted substitution

B. Fire Suppression Interconnect
   1. Provide as indicated on Drawings
   2. Water working pressure: 125 psi
   3. Combination air release and vacuum breaker valves: 4-inch and smaller
      a. Provide integral type that functions as both an air vacuum breaker and an air release valve
      b. Manufacturers:
         1) Val-Matic "Series 800"
         2) Crispin “SL(A) Series”
         3) Or accepted substitution

## 2.9 PRESSURE REDUCING VALVES

A. Quantity: 1
   1. Intake Screen Backwash Line
      a. Pilot adjustment range 30 psi to 300 psi
      b. 250 psi maximum inlet
      c. 80 psi maximum outlet

B. Manufacturers:
   1. Zurn

2. Watt
3. Or Engineer accepted substitution

C. Design
4. The Pressure Reducing Valve shall maintain a constant pressure regardless of changing flow rate and/or inlet pressure
5. The valve shall be hydraulically operated, single diaphragm-actuated, globe pattern
6. The valve shall consist of three major components: the body; the cover; and the internal trim assembly
7. The diaphragm assembly shall be fully guided at both ends by a bearing in the valve cover and in the valve seat
8. All valve components shall be accessible and serviceable without removing the valve from the pipeline
9. The control system shall consist of a 2-way adjustable pilot valve

## 2.10 PRESSURE REDUCING AND SUSTAINING VALVES WITH SOLENOID OVERRIDE

A. Quantity: 1
1. Fire Suppression Interconnect
a. Pilot adjustment range 25 psi to 200 psi
b. 250 psi maximum inlet
c. 125 psi maximum outlet

B. Manufacturers:
1. Nelson, Model 800 Series
2. Or Engineer accepted substitution

C. Design
3. The Pressure Reducing and Sustaining Valve shall maintain constant pressure regardless of changing flow rate and/or inlet pressure
4. The valve shall be hydraulically operated, single diaphragm-actuated, globe pattern
5. Valve shall be capable of leak-tight ON/OFF control by electrically-actuated solenoid
6. The valve shall consist of three major components: the body; the cover; and the internal trim assembly
7. The diaphragm assembly shall be fully guided at both ends by a bearing in the valve cover and in the valve seat
8. All valve components shall be accessible and serviceable without removing the valve from the pipeline
9. The control system shall consist of a 2-way adjustable pilot valve

# PART 3 EXECUTION

## 3.1 INSTALLATION

A. General:

1. Carefully inspect valve before installation. Clean interior. Operate valve to determine parts in proper working order, with valves seating and drain valve operating properly
2. Install valves and accessories in accordance with the manufacturer's recommendation and in accordance with referenced standards and specifications and shown on Drawings
3. Provide a union or flanged connection within 2 feet of each threaded end valve unless the valve can otherwise be easily removed from piping
4. Set valve level and plumb. Actuator to be set plumb or actuator can be set at 90 degrees to valve body if required for proper actuation
5. Install valve in proper orientation for flow and disc seating
6. Test valves installed with piping for hydrostatic testing, ensure each valve is operated and able to maintain test pressure and seal tight to isolate each piping section

B. Check valves: Install ball check valves in vertical piping

C. Air release and Vacuum-Relief Valves:
   1. Provide exhaust piping to a suitable disposal point

3.2 ADJUSTMENT

A. Check and adjust valves, actuators, pilotry, and accessories for smooth operation in accordance with manufacturer's instructions

3.3 VALVE SCHEDULE, VALVES LARGER THAN 4-INCHES

A. Install valves as scheduled in the Valve Schedule provided below
   1. FL: Flanged Joints
   2. C: Compression Fittings
   3. CV: Check Valve
   4. BFV: Butterfly Valve
   5. BV: Ball Valve
   6. GV: Gate Valve
   7. PR/PS: Pressure Relief and Pressure Sustaining Valve w/ Solenoid Override
   8. PV: Plug Valve
   9. ARV: Air/Vacuum Combination Relief Valve

| VALVE SCHEDULE | | | | | |
|---|---|---|---|---|---|
| **Type** | **Valve ID** | **Size** | **Area Location** | **Type of Installation** | **Valve End Connection** |
| PV | PV-1 | 6-inch | Suction Line – Vault Exterior | Outdoor Buried | MJ x MJ |
| PV | PV-2 | 4-inch | Suction Line – Pump 200 | Indoor | FL x FL |
| PV | PV-3 | 4-inch | Suction Line – Pump 210 | Indoor | FL x FL |
| CV | CV-1 | 4-inch | Discharge Line – Pump 200 | Indoor | FL x FL |
| CV | CV-2 | 4-inch | Discharge Line – Pump 210 | Indoor | FL x FL |
| BFV | BFV-1 | 4-inch | Discharge Line – Pump 200 | Indoor | FL x FL |
| BFV | BFV-2 | 4-inch | Discharge Line – Pump 210 | Indoor | FL x FL |
| BFV | BFV-3 | 2-inch | Hydropneumatic Tank Connection | Indoor | FL x FL |
| GV | GV-1 | 4-inch | Distribution Line to Irrigation | Outdoor Buried | MJ x MJ |

| | | | | | |
|---|---|---|---|---|---|
| GV | GV-2 | 6-inch | Fire Suppression Interconnection | Outdoor Buried | MJ x MJ |
| PR/PS | PR/PS-1 | 4-inch | Fire Suppression Interconnection | Indoor | FL x FL |
| CV | CV-3 | 4-inch | Fire Suppression Interconnection | Indoor | FL x FL |
| BFV | BFV-4 | 4-inch | Fire Suppression Interconnection | Indoor | FL x FL |
| ARV | ARV-1 | TBD | Fire Suppression Interconnection | Indoor | TBD |
| CV | CV-4 | 2-inch | VAF Filter Flush Line | Indoor | FL x FL |
| BV | BV-1 | 2-inch | VAF Filter Flush Line | Indoor | FL x FL |
| PRV | PRV-1 | 1-inch | Intake Screen Backwash Line | Indoor | TBD |
| BV | BV-2 | 1-inch | Intake Screen Backwash Line | Indoor | TBD |
| ARV | ARV-2 | TBD | Sump Pump Discharge Line | Indoor | TBD |
| | | | | | |

END OF SECTION

SECTION 15090

SUPPORTS AND ANCHORS

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Pipe and equipment hangers and supports

B. Equipment bases and supports

C. Flashing and sealing equipment

1.2 RELATED SECTIONS

A. Section 03000 – Concrete

B. Section 05501 – Anchor Bolts and Expansion Anchors

C. Section 05590 – Miscellaneous Metal

D. Section 09900 – Coatings

E. Section 15060 – Pipe and Pipe Fittings

1.3 REFERENCES

A. American Society of Testing and Materials (ASTM)
   1. F708 – Design and Installation of Rigid Pipe Hangers

B. International Building Code (IBC)

C. Manufacturers Standardization Society (MSS)
   1. SP58 – Pipe Hangers and Supports – Materials, Design and Manufacturer
   2. SP69 – Pipe Hangers and Supports – Selection and Application
   3. SP89 – Pipe Hangers and Supports – Fabrication and Installation Practices

1.4 SYSTEM DESCRIPTION

A. Provide complete system of pipe supports and anchors for all piping

B. Support all piping to prevent undue strain on any valve, fitting or piece of equipment

C. Provide support and anchors at changes in direction, change in elevation, adjacent to flexible couplings, and where valve actuators/positioners add additional weight and stresses

D. Do not install in equipment access areas

## 1.5 PERFORMANCE REQUIREMENTS

A. Pipe support system components to withstand total dead load
   1. Dead load: Weight of pipe filled with water plus any insulation
      a. Dead load defined as weight of pipe filled with water plus any insulation
   2. Factor of safety: 5 minimum

B. Pipe restraint system components to withstand thrusts created by fluid pressure within pipes
   1. Factor of safety: 5 minimum

C. Do not exceed manufacturers' recommended loads

D. Brace all piping 2½ inch and larger for Seismic Design Category B forces in accordance with International Building Code

## 1.6 SUBMITTALS

A. Shop drawings: Indicate system layout with location and detail of trapeze hangers. Include location of pipe supports

B. Product data: Provide manufacturers catalog data including load capacity

C. Design data: Indicate load carrying capacity of trapeze, multiple pipe, and riser support hangers

D. Manufacturer's installation instructions: Indicate special procedures and assembly of components

# PART 2 PRODUCTS

## 2.1 MATERIALS

A. Not submerged: Galvanized steel except where otherwise indicated or specified

B. Submerged: Stainless steel, 304L

C. Use stainless steel materials for stainless steel pipe or galvanized steel with non-conductive isolation gaskets to prevent electrolytic corrosion

## 2.2 PIPE HANGERS AND SUPPORTS

A. Manufacturers:
   1. Anvil International
   2. Unistrut Corporation
   3. Elcen Metal Products Co.
   4. PHD Manufacturing, Inc
   5. Or accepted substitution

B. Plumbing piping – water:
   1. Conform to requirements of Section 15060
   2. Hangers for pipe sizes ½ to 1½ inch: Stainless steel, adjustable swivel, split ring
   3. Multiple or trapeze hangers: Steel channels with welded spacers and hanger rods
   4. Wall support for pipe sizes to 3 inches: Cast iron hook
   5. Vertical support: Stainless steel riser clamp
   6. Copper pipe support: Stainless steel ring, adjustable, copper plated

2.3 ACCESSORIES

A. Hanger rods: ASTM A307, Grade R. Mild steel threaded both ends, threaded one end, or continuous threaded

B. Isolation pads: neoprene waffle type
   1. Manufacturers: Mason Industries "Type W", Fabreka Production Company "Fabcel", or accepted substitution

C. Insulation protection shield: ANSI/MSS SP-58

2.4 FINISHING

A. All hangers, rods, clamps, protective shields, metal framing support components, and hanger accessories: Galvanized Steel unless otherwise indicated or specified

B. For stainless steel components: none required

PART 3 EXECUTION

3.1 INSTALLATION

A. Install in accordance with manufacturer's instructions

B. Pipe supports and anchors are not indicated on drawings in all locations; supports are indicated for most larger pipes

C. Provide system of pipe supports and anchors for all piping

D. Provide additional supports as required adjacent to couplings

E. Pipe supports shall be provided at all changes in direction and adjacent to flexible couplings

3.2 PIPE HANGERS AND SUPPORTS

A. For designated pipe supports located adjacent to equipment and other locations requiring vibration isolation, provide isolation pad

B. Support horizontal piping as scheduled

C. Install hangers to provide minimum ½ inch space between finished covering and adjacent work

D. Brace all piping 2-1/2 inch and larger for Seismic Design Category B forces in accordance with local building codes

E. Place hangers within 12 inches of each horizontal elbow

F. Use hangers with 1½ inch minimum vertical adjustment

G. Where several pipes can be installed in parallel and at same elevation, provide multiple or trapeze hangers

H. Support riser piping independently of connected horizontal piping

I. Design hangers for pipe movement without disengagement of supported pipe

J. Piping connections to equipment: Provide pipe support adjacent to equipment. Pipes shall not be supported on equipment

K. Do not support lower pipe from pipe above

L. Support pipe 1½ inch minimum from walls and 3 inch minimum below ceilings

M. From framing strut systems, cut ends and paint exposed metal with galvanized paint

N. Prime coat exposed steel hangers and supports. Refer to Section 09900. Hangers and supports located in crawl spaces, pipe shafts, and suspended ceiling spaces are not considered exposed

O. Do not install pipe supports in equipment access areas

3.3 EQUIPMENT BASES AND SUPPORTS

A. Provide housekeeping pads of concrete, minimum 3 inches thick and extending 4 inches beyond supported equipment. Refer to Drawings for location of pads

B. Provide templates, anchor bolts, and accessories for mounting and anchoring equipment

C. Construct supports of steel members. Brace and fasten with flanges bolted to structure

D. Provide rigid anchors for pipes after vibration isolation components are installed

3.4 SCHEDULES

A. Maximum hanger spacing and rod size for cast iron, ductile iron, steel , HDPE and PVC:

| **Pipe size (inches)** | **Max. hanger spacing (feet)** | **Hanger rod diameter (inches)** |
|---|---|---|
| 1½ and smaller | 6 | 3/8 |
| 1 ½ to 2 | 8 | 3/8 to 1/2 |
| 2 ½ to 3 | 10 | 1/2 to 5/8 |
| 4 | 10 | 5/8 |
| 6 and 8 | 12 | 3/4 |
| 10 and 12 | 14 | 7/8 |
| 14 and16 | 16 | 1 |
| 18 | 16 | 1-1/8 |
| 20 and larger | 18 | 1-1/4 |
| PVC (all sizes) | 6 | Same as for steel |

B. Maximum hanger spacing and rod size for copper and stainless steel piping
   1. Spacing:
      a. 1 inch and smaller: 5 feet maximum
      b. 1¼ inch and larger: 2 feet less than maximum spacing for steel pipe of same size
   2. Hanger rod size: Same as for steel

END OF SECTION